THE J-0 PROPERTY ASCENDS ALONG INJECTIVE FINITE TYPE MAPS WITH SEPARABLE FRACTION FIELD EXTENSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A finite type map from a Noetherian J-0 domain is smooth at the generic point when the fraction field extension is separable, and regularity ascends along smooth maps. Algebra, Lemma \ref{algebra-lemma-smooth-at-generic-point} and Algebra, Lemma \ref{algebra-lemma-regular-goes-up}, cited at proof lines 11681-11683 and 11685-11686. Give the smoothness at the generic point and regularity of the target.

Source reference. Original lemma 07PB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9791–9791. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07ND: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9758–9758. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9760–9760. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to S$ be a ring map. Assume that*

(1) *R is a Noetherian domain and J-0,*

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(2) *$R \to S$ is injective and of finite type, and*
(3) *S is a domain, and*
(4) *the induced extension of fraction fields is separable.*

Then S is J-0.

Proof. We may replace R by a principal localization and assume R is a regular ring. By Algebra, Lemma 07ND the ring map $R \to S$ is smooth at (0). Hence after replacing S by a principal localization we may assume that S is smooth over R. Then S is regular too, see Algebra, Lemma 07NF. □